Radiology

ORIGINAL RESEARCH • COMPUTER APPLICATIONS

Automated Comprehensive CT Assessment of the Risk of Diabetes and Associated Cardiometabolic Conditions

Yoosoo Chang, MD, PhD • Soon Ho Yoon, MD, PhD* • Ria Kwon, PhD • Jeonggyu Kang, MD • Young Hwan Kim, MD, PhD • Jong-Min Kim, PhD • Han-Jae Chung, PhDc • JunHyeok Choi, PhDc • Hyun-Suk Jung, MD, PhD • Ga-Young Lim, PhD • Jiin Ahn, MSPH • Sarah H. Wild, MB, BChir, PhD • Christopher D. Byrne, MBBCh, PhD • Seungho Ryu, MD, PhD*

From the Center for Cohort Studies (Y.C., R.K., J.K., J.H.C., H.S.J., G.Y.L., J.A., S.R.) and Department of Occupational and Environmental Medicine (Y.C., S.R.), Kangbuk Samsung Hospital, Sungkyunkwan University School of Medicine, 29 Saemunan-ro, Jongno-gu, Seoul 03181, Republic of Korea; Department of Clinical Research Design & Evaluation, Samsung Advanced Institute for Health Sciences & Technology, Sungkyunkwan University, Seoul, Republic of Korea (Y.C., S.R.); Department of Radiology, Seoul National University Hospital, Seoul National College of Medicine, Seoul, Republic of Korea (S.H.Y.); Research & Science Division, MEDICAL IP, Seoul, Republic of Korea (J.M.K., H.J.C.); Institute of Medical Research, Sungkyunkwan University School of Medicine, Suwon, Republic of Korea (R.K., G.Y.L.); Department of Nuclear Medicine, Kangbuk Samsung Hospital, Sungkyunkwan University School of Medicine, Seoul, Republic of Korea (Y.H.K.); Usher Institute, University of Edinburgh, Edinburgh, United Kingdom (S.H.W.); Department of Nutrition and Metabolism, University of Southampton Faculty of Medicine, Southampton, United Kingdom (C.D.B.); and National Institute for Health and Care Research Southampton Biomedical Research Centre, University Hospital Southampton, Southampton, United Kingdom (C.D.B.). Received December 20, 2023; revision requested February 6, 2024; final revision received May 8; accepted May 22. **Address correspondence to** S.R. (email: *sh703.yoo@gmail.com*).

Supported by SKKU Excellence in Research Award Research Fund, Sungkyunkwan University (2022), and the National Research Foundation of Korea, Ministry of Science ICT and Future Planning (NRF-2021R1A2C1012626). C.D.B. was funded in part by the National Institute for Health Research Southampton Biomedical Research Centre (NIHR 203319). The funding agency of the study had no role in the study design, data collection, data analysis, data interpretation, or writing of the manuscript.

Conflicts of interest are listed at the end of this article.

* Y.C. and S.H.Y. contributed equally to this work.

See also the editorial by Pickhardt in this issue.

Radiology 2024; 312(2):e233410 • https://doi.org/10.1148/radiol.233410 • Content codes: AI GI CT

Background: CT performed for various clinical indications has the potential to predict cardiometabolic diseases. However, the predictive ability of individual CT parameters remains underexplored.

Purpose: To evaluate the ability of automated CT-derived markers to predict diabetes and associated cardiometabolic comorbidities.

Materials and Methods: This retrospective study included Korean adults (age ≥ 25 years) who underwent health screening with fluorine 18 fluorodeoxyglucose PET/CT between January 2012 and December 2015. Fully automated CT markers included visceral and subcutaneous fat, muscle, bone density, liver fat, all normalized to height (in meters squared), and aortic calcification. Predictive performance was assessed with area under the receiver operating characteristic curve (AUC) and Harrell C-index in the cross-sectional and survival analyses, respectively.

Results: The cross-sectional and cohort analyses included 32166 (mean age, 45 years ± 6 [SD], 28833 men) and 27298 adults (mean age, 44 years ± 5 [SD], 24820 men), respectively. Diabetes prevalence and incidence was 6% at baseline and 9% during the 7.3-year median follow-up, respectively. Visceral fat index showed the highest predictive performance for prevalent and incident diabetes, yielding AUC of 0.70 (95% CI: 0.68, 0.71) for men and 0.82 (95% CI: 0.78, 0.85) for women and C-index of 0.68 (95% CI: 0.67, 0.69) for men and 0.82 (95% CI: 0.77, 0.86) for women, respectively. Combining visceral fat, muscle area, liver fat fraction, and aortic calcification improved predictive performance, yielding C-indexes of 0.69 (95% CI: 0.68, 0.71) for men and 0.83 (95% CI: 0.78, 0.87) for women. The AUC for visceral fat index in identifying metabolic syndrome was 0.81 (95% CI: 0.80, 0.81) for men and 0.90 (95% CI: 0.88, 0.91) for women. CT-derived markers also identified US-diagnosed fatty liver, coronary artery calcium scores greater than 100, sarcopenia, and osteoporosis, with AUCs ranging from 0.80 to 0.95.

Conclusion: Automated multiorgan CT analysis identified individuals at high risk of diabetes and other cardiometabolic comorbidities.

Supplemental material is available for this article.

The application of machine and deep learning algorithms has revolutionized the landscape of body composition analysis from images, rendering it less labor intensive and less dependent on manual intervention (1,2). However, lack of systematic integration has hindered the full utilization of these algorithms in clinical practice (1).

Data from opportunistic use of CT imaging beyond its primary clinical indication have shown promise in incidental osteoporosis screening (3,4) and in quantifying aortic calcification, visceral and subcutaneous fat, muscle mass, and liver fat content (5–8). Even a single CT image at the level of the third lumbar vertebra can provide precise information regarding visceral and subcutaneous fat, muscle mass, and bone density (9). CT scans can potentially predict cardiovascular disease events and all-cause mortality, raising the prospect of tangible benefits of CT in patient risk stratification (10–12).

However, the predictive ability of individual imaging parameters for cardiometabolic diseases, traditionally diagnosed using conventional modalities, remains

Abbreviations

AUC = area under the receiver operating characteristic curve, BMI = body mass index, CAC = coronary artery calcium, DM = diabetes mellitus, FDG = fluorodeoxyglucose, PDFF = proton density fat fraction

Summary

In a cohort of Korean adults who underwent health screening with fluorine 18 fluorodeoxyglucose PET/CT, automated multiorgan CT analysis identified individuals at risk for type 2 diabetes and associated cardiometabolic comorbidities.

Key Results

- In a retrospective study of 32 166 Korean adults who underwent health screening including fluorine 18 fluorodeoxyglucose PET/CT, diabetes prevalence and incidence was 6% at baseline and 9% during the follow-up, respectively.
- Automated CT-derived markers predicted new-onset diabetes, with Harrell C-index of 0.68 for men and 0.82 for women.
- Automated CT-derived markers identified fatty liver, metabolic syndrome, coronary artery calcium scores greater than 100, sarcopenia, and osteoporosis (areas under the receiver operating characteristic curve, 0.80–0.95).

underexplored. Type 2 diabetes mellitus (DM) is a common metabolic disorder associated with considerable comorbidities and complications and is frequently diagnosed late (13). Body composition, including muscle mass and differential fat distribution, has the potential to predict type 2 DM and related complications (14,15).

The aim of this study was to evaluate the ability of fully automated CT-derived markers for identifying prevalent and incident diabetes and associated cardiometabolic comorbidities (metabolic syndrome, sarcopenia, osteoporosis, fatty liver, and coronary artery calcium [CAC]) among Korean adults participating in a health screening program (Fig S1).

Materials and Methods

This retrospective study was approved by the Institutional Ethics Committee (institutional review board no. KBSMC 2022–04–028), and the requirement for informed consent was waived.

Patients

In this study, a subset of patients from the Kangbuk Samsung Health Study, a prospective cohort study involving Korean adults who are primarily company employees and their spouses and who are undergoing health screening as per South Korea's Industrial Safety and Health Law was analyzed. Inclusion criteria included those who underwent fluorine 18 (^{18}F) fluorodeoxyglucose (FDG) PET/CT between 2012 and 2015 as part of a comprehensive health examination (see Appendix S1 for exclusion criteria details). Despite debates over the role of ^{18}F-FDG PET/CT in health screening due to radiation exposure and costs, this examination is performed for cancer screening in Korea.

Measurements

PET/CT image acquisition.—After a minimum 8-hour fast, torso ^{18}F-FDG PET/CT scans were obtained using a PET/CT system (Discovery 600; GE HealthCare) without the use of contrast materials. Information on imaging acquisition parameters is presented in Appendix S1.

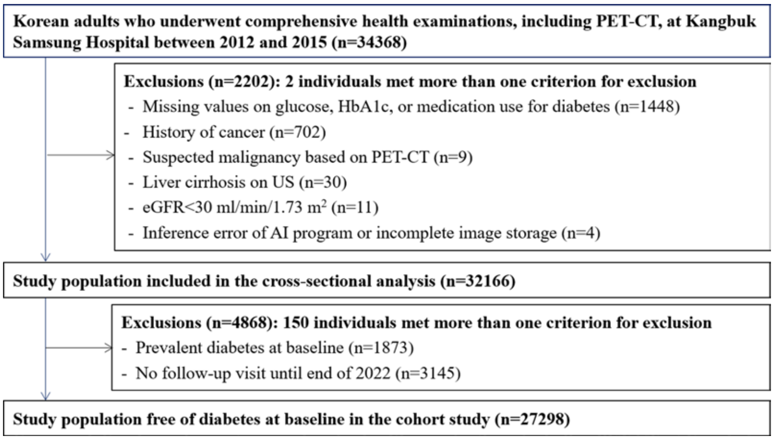
Body composition, liver, and abdominal aorta calcification analysis.—Noncontrast torso CT images from the PET/CT acquisition were processed using U.S. Food and Drug Administration–approved commercial software (version 1.2.0.0, DeepCatch, Medical IP; *http://www.medicalip.com*) (Appendix S1, Fig S2) to measure the sectional areas (in centimeters squared) of skeletal muscle, subcutaneous fat, and visceral fat, normalized to height (in meters squared), as indexes. This study focused on these areas, as well as the visceral-to-subcutaneous fat ratio, and the trabecular density at the L3 level, given their strong predictive value for overall mortality (16). Additionally, the software automatically calculated the volumetric liver density and estimated MR proton density fat fraction (PDFF) values of the liver based on a deep learning–based image synthesis of MRI PDFF from CT images (Appendix S1). Aortic calcification was calculated based on the Agatston calcium score for all aortic regions on the patient's CT images.

Definition of diabetes and other variables.—Data encompassing physical measurements, abdominal US findings, and serum biochemical measurements were systematically collected before ^{18}F-FDG PET/CT imaging was performed as part of the health screening program. Demographic characteristics, health behaviors (smoking, alcohol consumption, and physical activity), medical history, and medication use were assessed by using standardized self-administered questionnaires (Appendix S1).

Blood samples obtained after 10 hours of fasting included measurements of glucose level, hemoglobin A_{1c} level, and lipid profiles. Type 2 DM was defined as having one or more of the following: a fasting serum glucose level of 126 mg/dL (7.0 mmol/L) or greater, hemoglobin A_{1c} level 6.5% or greater (≥ 48 mmol/mol), or current use of insulin or glucose-lowering medications for diabetes management. Metabolic syndrome and sarcopenia, assessed with impedance analysis, were defined using standard criteria (Appendix S1) (17–19). The CAC score based on CAC CT was categorized as 0–100 or greater than 100, with scores above 100 indicating a critical threshold for statin eligibility (Appendix S1) (20,21).

Statistical Analysis

Descriptive statistics, including means ± SDs, percentages, and medians and IQRs as appropriate, were used to summarize patient characteristics in the cross-sectional study based on the prevalence of type 2 DM and the cohort study based on type 2 DM development among initially diabetes-free individuals. Robust Poisson regression models (22,23) were used to estimate the prevalence ratios (and 95% CIs) for type 2 DM and each clinical disease by comparing parameter quartiles with the lowest quartile as the reference, while Cox proportional hazard models were used to determine the hazard ratios (and 95% CIs) for diabetes onset by comparing each of the three other quartiles of each parameter against the reference category of the lowest quartile (see details in Appendix S1). To assess the relationship between



Flowchart of the selection of the study participants. AI = artificial intelligence, eGFR = estimated glomerular filtration rate, HbA1c = glycated hemoglobin.

CT-derived markers and diabetes risk, restricted cubic splines with knots at the 5th, 27.5th, 50th, 72.5th, and 95th percentiles of the sample distribution were used to provide a flexible estimate of the concentration–response relationship between CT-derived markers and diabetes risk.

The performance of the imaging parameters in predicting diabetes and related cardiometabolic diseases relative to standard or commonly used practical measures was assessed by using the area under the receiver operating characteristic curve (AUC). For the combination model incorporating multiple CT-derived measures, predicted probabilities for diabetes were generated using software and the postestimation command "predict" after fitting multivariable logistic regression models. Differences in the AUC between anthropometric measures and CT-derived measures were evaluated with the Stata (StataCorp) "roccomp" command. The optimal model was determined as a combination of CT-derived measures that maximized the AUC. In this cohort study, the predictive capabilities of conventional measures and CT-derived imaging parameters were compared using the Harrell C-index, a measure adapted for survival analysis (24). The predictive values of CT-derived markers were compared with those of conventional risk factor models, such as the American Diabetes Association diabetes risk scores and Leicester Diabetes Risk Scores. To address the issue of multiple testing, Bonferroni adjustment (a method commonly used for multiple testing correction) was applied. This study used routinely collected health screening data, with the sample size determined by enrolled patients during the study period. Statistical analyses were performed by two authors (Y.C. and S.R.) using Stata software (version 17.0; StataCorp), with statistical significance defined as $P < .05$.

Results

Patient Characteristics

Among 34368 patients who underwent ^{18}F-FDG PET/CT between 2012 and 2015, 2202 patients were excluded due to missing diabetes-related data, incomplete image storage, cancer history, liver cirrhosis, and low estimated glomerular filtration rate, resulting in 32166 patients for the cross-sectional study (Figure 1, Appendix S1). In the cohort study, 27298 individuals who initially did not have diabetes were followed up until December 31, 2022, for incident diabetes.

The cross-sectional study included 32166 patients (mean age, 45 years ± 6 [SD], 28833 men and 3333 women) (Table 1). At baseline, the overall prevalence of type 2 DM was 6%: 6% in men and 4% in women. Individuals without type 2 DM were younger than those with diabetes (mean, 44 years ± 5 vs 47 years ± 6 [$P < .001$] for men and 47 years ± 9 vs 57 years ± 10 [$P < .001$] for women) and were less likely to have hypertension (17% vs 40% [$P < .001$] for men and 11% vs 39% [$P < .001$] for women) and use lipid-lowering medications (4% vs 23% [$P < .001$] for men and 5% vs 29% [$P < .001$] for women). Individuals without type 2 DM also had a lower body mass index (BMI), waist circumference, and impedance-derived fat index ($P < .001$).

The type 2 DM group exhibited increased impedance-based skeletal muscle index and CT-derived L3 muscle area indexes but decreased muscle density. Initially, the muscle area index was higher in individuals with diabetes, but the difference did not persist after adjusting for age and BMI. The age- and BMI-adjusted mean muscle area index was 52.9 (95% CI: 52.8, 52.9) for men without diabetes and 52.2 (95% CI: 52.0, 52.4) for men with diabetes. For women, the adjusted means showed negligible differences between those without diabetes (39.8 [95% CI: 39.6, 39.9] and those with diabetes (40.0 [95% CI: 39.4, 40.7]). The diabetes group had higher subcutaneous fat and visceral fat indexes, a greater visceral-to-subcutaneous fat ratio, lower liver densities, and more aortic calcification ($P < .001$). Tables S1–S3 provide age- and sex-specific distributions of CT-derived parameters. The cohort study included 27298 (mean age, 43.8 years ± 4.8, 24820 men) individuals (Table S4). The pattern in the difference between individuals without and individuals with incident diabetes was similar to that between those without and those with prevalent diabetes across anthropometric, impedance, and CT-derived measures.

Relationship of CT-derived Parameters to Prevalent Diabetes and Other Cardiometabolic Conditions

Table 2 displays the AUCs for both conventional and CT-derived parameters for their association with prevalent diabetes in men and women. In men, the CT-derived L3 visceral fat index, liver PDFF (or liver density), and aortic calcification, which are individual markers, demonstrated higher AUC values for prevalent diabetes than BMI ($P < .001$). In women, while all three parameters exhibited higher AUC values, only the visceral fat index and aortic calcification showed differences ($P < .001$). The combined use of visceral fat area, subcutaneous fat, liver PDFF, and aortic calcification yielded the highest AUC for the prevalence of diabetes in both men and women. Specifically, the AUC for this combination was 0.75 (95% CI: 0.74, 0.77) for men and 0.85 (95% CI: 0.82, 0.89) for women. After adjusting

Table 1: Baseline Characteristics of the Participants by Prevalent Diabetes

Characteristic	Men without Diabetes (n = 27090)	Men with Diabetes (n = 1743)	Women without Diabetes (n = 3203)	Women with Diabetes (n = 130)	*P* Value for without vs with Diabetes: Men	*P* Value for without vs with Diabetes: Women
Age (y)*	44.1 ± 5.0	47.3 ± 6.3	46.6 ± 8.6	57.1 ± 9.8	<.001	<.001
Age range (y)	27–80	37–77	29–83	39–78		
Current smoker	8761 (33.4)	628 (37.9)	75 (2.7)	3 (2.9)	<.001	.89
Alcohol user	8656 (32.8)	696 (41.5)	179 (6.4)	4 (4.1)	<.001	.36
HEPA	3788 (14.2)	293 (17.2)	383 (12.3)	31 (24.4)	<.001	<.001
Lipid-lowering drugs	1111 (4.1)	403 (23.3)	157 (4.9)	37 (28.7)	<.001	<.001
Hypertension	4667 (17.2)	690 (39.6)	347 (10.8)	51 (39.2)	<.001	<.001
Metabolic syndrome	7095 (26.2)	1180 (67.7)	215 (6.7)	55 (42.3)	<.001	<.001
Anthropometry						
BMI (kg/m^2)*	24.5 ± 2.7	25.9 ± 3.2	22.3 ± 3.0	24.7 ± 4.1	<.001	<.001
Waist circumference (cm)*	86.1 ± 7.3	90.2 ± 8.1	76.9 ± 8.2	83.9 ± 10.6	<.001	<.001
Impedance analysis						
Fat percentage*	23.1 ± 4.9	25.5 ± 5.3	30.5 ± 5.9	33.9 ± 6.3	<.001	<.001
Fat mass per height^2†	5.6 (4.5–6.7)	6.4 (5.3–7.8)	6.6 (5.4–8.2)	8.1 (6.6–10.0)	<.001	<.001
Skeletal muscle mass (kg)†	31.3 (29.0–33.6)	31.8 (29.4–34.3)	20.9 (19.4–22.5)	20.6 (18.8–23.0)	<.001	.45
ASM (kg)†	23.8 (22.0–25.6)	24.1 (22.1–25.9)	15.6 (14.4–17.0)	15.1 (13.5–17.3)	<.001	.52
SMI (ASM/height^2)†	8.0 (7.6–8.4)	8.1 (7.7–8.6)	6.1 (5.8–6.5)	6.3 (5.8–6.8)	<.001	<.001
CT-derived measures						
Muscle area index at L3†	52.3 (47.9–57.2)	54.3 (49.4–59.5)	39.2 (36.2–42.7)	42.1 (38.7–46.9)	<.001	<.001
Muscle density (HU)†	45.1 (42.1–48.1)	44.2 (40.8–47.3)	38.3 (34.1–41.8)	34.5 (30.1–37.7)	<.001	<.001
Visceral fat index at L3†	41.0 (29.4–53.4)	54.3 (42.3–68.0)	17.7 (9.0–29.9)	42.7 (30.3–54.8)	<.001	<.001
Visceral fat density (HU)†	−92.4 (−96.0 to 88.5)	−92.5 (−95.8 to 89.3)	−92.2 (−96.6 to 87.1)	−94.1 (−98.2 to 89.8)	.002	.008
SC fat index at L3†	42.8 (33.6–53.6)	45.0 (35.2–56.8)	56.9 (43.0–73.2)	69.7 (54.4–89.0)	<.001	<.001
SC fat density (HU)†	−91.6 (−95.3 to 87.5)	−89.8 (−93.6 to 85.7)	−96.3 (−99.7 to 92.2)	−96.3 (−99.1 to 91.7)	<.001	.83
VS ratio†	0.9 (0.7–1.2)	1.2 (0.9–1.5)	0.3 (0.2–0.4)	0.6 (0.4–0.8)	<.001	<.001
Vertebral density (HU)†	272.3 (243.9–303.7)	269.9 (238.5–302.2)	299.4 (256.5–337.9)	249.1 (210.0–291.5)	.003	<.001
Liver density (HU)†	55.4 (50.7–59.0)	50.6 (42.9–55.5)	55.8 (52.6–58.8)	51.4 (43.5–55.9)	<.001	<.001
Liver PDFF†	6.8 (5.5–8.7)	8.7 (6.8–12.3)	6.2 (5.2–7.5)	8.2 (6.4–12.2)	<.001	<.001
Aortic calcification (Agatston score)†	9.5 (2.9–48.6)	57.2 (7.6–327.1)	5.7 (1.9–21.9)	122.5 (14.3–636.1)	<.001	<.001

Note.—Except where noted, data are numbers of patients, with percentages in parentheses. Alcohol users refers to individuals who consume ≥ 20 g of ethanol per day. HEPA refers to engaging in either *(a)* vigorous-intensity activities for at least 3 days per week, totaling 1500 or more metabolic equivalent task (MET) minutes per week, or *(b)* a combination of walking, moderate-intensity, or vigorous-intensity activities across 7 days, achieving a minimum of 3000 MET minutes per week. ASM = appendicular skeletal muscle mass, BMI = body mass index (calculated as weight in kilograms divided by height in meters squared), HEPA = health-enhancing physical activity, HU = Hounsfield unit, L3 = third lumbar vertebra, PDFF = proton density fat fraction, SC = subcutaneous, SMI = skeletal muscle index, VS = visceral to subcutaneous fat.

* Normally distributed continuous variables as expressed as means ± SDs.

† Nonnormally distributed continuous variables expressed as medians, with IQRs in percentages.

for age, center, and year of examination, the prevalence ratios (and 95% CIs) for diabetes, comparing the quartiles with the lowest quartile (reference), showed distinct patterns for men and women (Table S5). In men, the highest prevalence ratio was the visceral fat, followed by liver PDFF. Conversely, in women, the highest prevalence ratio was observed for the visceral fat index, followed by the visceral-to-subcutaneous fat ratio, aortic calcification, and liver PDFF. The CT-visceral fat index consistently outperformed the impedance-derived body composition indexes (skeletal muscle index, fat mass index, and body fat percentage) in predicting prevalent diabetes (Table S6).

Table 3 highlights the discrimination performance of the CT-derived markers in identifying various comorbidities using clinical standards for comparison. Liver PDFF was a reliable measure for identifying US-diagnosed fatty liver, with an AUC of 0.81 (95% CI: 0.80, 0.81) for men and 0.80 (95% CI: 0.78, 0.82) for women. Aortic calcification demonstrated a high AUC in identifying CAC scores greater than 100, particularly in women, where the AUC

Table 2: Discrimination Performance of CT-derived Parameters in Identifying Prevalent Diabetes

	Men (*n* = 28833)			Women (*n* = 3333)		
Variable	AUC*	*P* Value for Reference vs Listed Variable	*P* Value for Reference vs Listed Variable	AUC*	*P* Value for Reference vs Listed Variable	*P* Value for Reference vs Listed Variable
Anthropometric measures						
Body mass index	0.64 (0.62, 0.65)	Reference		0.68 (0.64, 0.73)	Reference	
Waist circumference	0.65 (0.63, 0.66)	.01	Reference	0.71 (0.67, 0.76)	.03	Reference
Impedance analysis						
Body fat percentage	0.63 (0.61, 0.64)	.16	.002	0.66 (0.61, 0.71)	.13	.003
Fat mass per height^2	0.64 (0.63, 0.65)	.27	.20	0.68 (0.63, 0.72)	.48	.01
SMI (ASM/height^2)	0.57 (0.55, 0.58)	<.001†	<.001†	0.57 (0.50, 0.63)	<.001†	<.001†
CT-derived measures						
Muscle area index at L3	0.58 (0.56, 0.59)	<.001†	<.001†	0.65 (0.60, 0.70)	.06	.006
Visceral fat index at L3	0.70 (0.68, 0.71)	<.001†	<.001†	0.82 (0.78, 0.85)	<.001†	<.001†
Subcutaneous fat index at L3	0.54 (0.53, 0.56)	<.001†	<.001†	0.66 (0.62, 0.70)	.13	.002
VS ratio	0.67 (0.66, 0.69)	<.001†	.003	0.80 (0.76, 0.84)	<.001†	<.001†
Muscle density (HU)	0.56 (0.55, 0.57)	<.001†	<.001†	0.68 (0.64, 0.73)	.94	.27
Visceral fat density (HU)	0.51 (0.50, 0.53)	<.001†	<.001†	0.58 (0.53, 0.63)	.001	<.001†
Subcutaneous fat density (HU)	0.58 (0.57, 0.60)	<.001†	<.001†	0.51 (0.47, 0.56)	<.001†	<.001†
Vertebral density (HU)	0.52 (0.51, 0.54)	<.001†	<.001†	0.70 (0.66, 0.75)	.63	.72
Liver PDFF	0.68 (0.66, 0.69)	<.001†	<.001†	0.73 (0.68, 0.77)	.11	.64
Liver density (HU)	0.68 (0.66, 0.69)	<.001†	<.001†	0.72 (0.67, 0.77)	.24	.88
Aortic calcification	0.67 (0.66, 0.69)	<.001†	.008	0.78 (0.74, 0.82)	.003	.03
Combination of CT-derived measures						
Visceral fat index + subcutaneous fat index	0.70 (0.69, 0.72)	<.001†	<.001†	0.82 (0.79, 0.86)	<.001†	<.001†
Visceral fat index + muscle area index	0.70 (0.68, 0.71)	<.001†	<.001†	0.82 (0.78, 0.85)	<.001†	<.001†
Visceral fat index + visceral fat density	0.70 (0.69, 0.72)	<.001†	<.001†	0.83 (0.79, 0.86)	<.001†	<.001†
Visceral fat index + liver PDFF	0.72 (0.71, 0.73)	<.001†	<.001†	0.82 (0.79, 0.86)	<.001†	<.001†
Visceral fat index + liver density	0.71 (0.70, 0.73)	<.001†	<.001†	0.82 (0.78, 0.85)	<.001†	<.001†
Visceral fat index + aortic calcification	0.73 (0.72, 0.74)	<.001†	<.001†	0.83 (0.80, 0.87)	<.001†	<.001†
Visceral fat index + aortic calcification + liver PDFF	0.75 (0.74, 0.76)	<.001†	<.001†	0.84 (0.81, 0.87)	<.001†	<.001†
Visceral fat index + aortic calcification + liver PDFF + subcutaneous fat index	0.75 (0.74–0.77)	<.001†	<.001†	0.85 (0.82, 0.89)	<.001†	<.001†

Note.—To address the issue of multiple testing, statistical significance was assessed using the Bonferroni adjustment for the 45 separate tests conducted for each sex, with a significance threshold set at α/45 (α divided by 45, 0.001 instead of 0.05). CT-derived measures, including muscle area, visceral fat area, and subcutaneous fat area at the L3 level, were normalized by dividing each individual area by the square of height. Body fat percentage was calculated using fat mass multiplied by 100 and then divided by the total body weight in kilograms. ASM = appendicular skeletal muscle mass, AUC = area under the receiver operating characteristic curve, BMI = body mass index, HU = Hounsfield unit, L3 = third lumbar vertebra, PDFF = proton density fat fraction, SMI = skeletal muscle index, VS = visceral to subcutaneous.

* Data in parentheses are 95% CIs.

† *P* value remained significant after Bonferroni correction.

Table 3: Discrimination Performance of CT-derived Parameters in Identifying Other Cardiometabolic Conditions

	AUC*	
Variable	Men (n = 28833)	Women (n = 3333)
Fatty liver based on US		
Available US data/fatty liver†	28824/13563	3332/614
Liver PDFF	0.81 (0.80, 0.81)	0.80 (0.78, 0.82)
Liver density (HU)	0.81 (0.80, 0.81)	0.79 (0.77, 0.81)
Coronary artery calcium (> 100)		
Available CAC CT data/CAC > 100†	4515/164	263/8
Aortic calcification	0.84 (0.80, 0.87)	0.95 (0.89, 1.00)
Sarcopenia		
Impedance data/sarcopenia†	27517/591	3034/21
Muscle area index at L3	0.90 (0.89, 0.91)	0.88 (0.83, 0.94)
Osteoporosis		
Available DXA data/osteoporosis†	819/66	630/60
Vertebral density (HU)	0.91 (0.85, 0.96)	0.92 (0.88, 0.95)
Metabolic syndrome		
Available components/metabolic syndrome†	28831/8275	3333/270
Muscle area index at L3	0.67 (0.66, 0.68)	0.77 (0.74, 0.80)
Visceral fat index at L3	0.81 (0.80, 0.81)	0.90 (0.88, 0.91)
Subcutaneous fat index at L3	0.71 (0.70, 0.71)	0.80 (0.77, 0.82
Visceral-to-subcutaneous fat ratio	0.65 (0.65, 0.66)	0.82 (0.80, 0.84)
Muscle density (HU)	0.54 (0.53, 0.55)	0.70 (0.66, 0.73)
Visceral fat density (HU)	0.57 (0.56, 0.58)	0.67 (0.64, 0.69)
Subcutaneous fat density (HU)	0.51 (0.50, 0.52)	0.57 (0.54, 0.60)
Vertebral density (HU)	0.52 (0.51, 0.53)	0.67 (0.64, 0.71)
Liver PDFF	0.64 (0.64, 0.65)	0.75 (0.72, 0.78)
Liver density (HU)	0.64 (0.64, 0.65)	0.73 (0.70, 0.77)
Aortic calcification	0.58 (0.57, 0.59)	0.73 (0.69, 0.76)

Note.—The L3 level sectional areas (in centimeters squared) of skeletal muscle were used for the analysis and subsequently normalized to height (in meters squared), referred to as the muscle area index. AUC = area under the receiver operating characteristic curve, CAC = coronary artery calcium, DXA = dual-energy x-ray absorptiometry, HU = Hounsfield unit, L3 = third lumbar vertebra, PDFF = proton density fat fraction.

* Data in parentheses are 95% CIs.

† The available sample size relevant to each reported outcome and the corresponding number of individuals with each condition.

reached 0.95 (95% CI: 0.89, 1.00). The L3 muscle area index effectively identified sarcopenia via the impedance-based skeletal muscle index, with an AUC of 0.90 (95% CI: 0.89, 0.91) for men and 0.88 (95% CI: 0.83, 0.94) for women. In the identification of a T score below −2.5 using spine dual-energy x-ray absorptiometry, L3 trabecular density exhibited AUC values exceeding 0.9 for both sexes. For identifying metabolic syndrome, the visceral fat index had a high AUC of 0.81 (95% CI: 0.80, 0.81) for men and 0.90 (95% CI: 0.88, 0.91) for women.

Relationship of CT-derived Parameters and the Development of Incident Diabetes

Over 183 651 person-years of follow-up (median follow-up of 7.3 years, up to 10.8 years), 2456 of 27 298 participants who were initially without diabetes developed incident type 2 DM, with an overall incidence rate of 13.4 (95% CI: 12.9, 13.9) per 1000 person-years (5.4 [95% CI: 4.4, 6.7] for women and 14.1 [95% CI: 13.6, 14.7] for men). The visceral fat index was the best predictive single imaging marker for incident type 2 DM in both sexes, surpassing conventional measures, impedance-derived body composition indexes, and clinical models, such as the American Diabetes Association and Leicester UK diabetes risk models (Tables 4, S7). Combining imaging markers increased the AUC for type 2 DM prediction, with the highest AUC for a combination of the visceral fat index, muscle area index, liver PDFF, and aortic calcification (0.69 [95% CI: 0.68, 0.71] for men and 0.83[95% CI: 0.78, 0.87] for women). After adjusting for age, center, and examination year, the highest hazard ratios were noted for the visceral fat index. The corresponding multivariable-adjusted hazard ratios for incident type 2 DM, comparing the highest quartile with the lowest quartile as a reference category, were 5.19 (95% CI: 4.52, 5.96) for men and 44.12 (95% CI: 10.58, 184.0) for women (Tables S8, S9). In spline regression analyses, the risk of diabetes exhibited the most pronounced increase across the spectrum of the visceral fat indexes for men (Fig S3). For women, diabetes risk sharply increased at values below 20, followed by a steady increase thereafter (Fig S4).

Table 4: Predictive Abilities of Various Measures for Identifying Incident Diabetes

	Men (n = 24820)			Women (n = 2478)		
Variable	Harrell C-Index	P Value*	P Value†	Harrell C-Index	P Value*	P Value†
Clinical prediction model						
ADA score	0.64 (0.63, 0.65)	<.001‡		0.80 (0.75, 0.85)	Reference	
Leicester UK diabetes risk	0.65 (0.64, 0.67)	Reference		0.78 (0.73, 0.83)	.30	
Anthropometric measures						
Body mass index	0.66 (0.65, 0.67)	.56	Reference	0.77 (0.72, 0.82)	.34	.73
Waist circumference	0.65 (0.64, 0.66)	.47	.06	0.77 (0.73, 0.82)	.38	Reference
Impedance analysis						
Body fat percentage	0.63 (0.61, 0.64)	<.001‡	<.001‡	0.74 (0.68, 0.80)	.04	.03
Fat mass per height^2	0.65 (0.63, 0.66)	.10	<.001‡	0.76 (0.70, 0.82)	.21	.37
SMI (ASM/height^2)	0.61 (0.60, 0.62)	<.001‡	<.001‡	0.69 (0.62, 0.75)	.002	.005
AI-automated CT measures						
Muscle area index at L3	0.61 (0.60, 0.62)	<.001‡	<.001‡	0.72 (0.67, 0.78)	.03	.04
Visceral fat index at L3	0.68 (0.67, 0.69)	<.001‡	<.001‡	0.82 (0.77, 0.86)	.43	.02
Subcutaneous fat index at L3	0.59 (0.58, 0.60)	<.001‡	<.001‡	0.76 (0.70, 0.81)	.11	.23
Visceral-to-subcutaneous fat ratio	0.61 (0.60, 0.62)	<.001‡	<.001‡	0.75 (0.71, 0.80)	.17	.44
Muscle density (HU)	0.51 (0.50, 0.52)	<.001‡	<.001‡	0.63 (0.57, 0.70)	<.001‡	<.001‡
Visceral fat density (HU)	0.54 (0.53, 0.56)	<.001‡	<.001‡	0.65 (0.59, 0.70)	<.001‡	<.001‡
Subcutaneous fat density (HU)	0.52 (0.51, 0.53)	<.001‡	<.001‡	0.57 (0.51, 0.63)	<.001‡	<.001‡
Vertebral density (HU)	0.50 (0.49, 0.51)	<.001‡	<.001‡	0.55 (0.48, 0.62)	<.001‡	<.001‡
Liver PDFF	0.63 (0.61, 0.64)	<.001‡	<.001‡	0.67 (0.60, 0.74)	.001	.003
Liver density (HU)	0.63 (0.61, 0.64)	<.001‡	<.001‡	0.65 (0.58, 0.72)	<.001‡	.001
Aortic calcification	0.56 (0.55, 0.58)	<.001‡	<.001‡	0.60 (0.53, 0.67)	<.001‡	<.001‡
Combination of AI-automated CT measures						
Visceral fat index + subcutaneous fat index	0.68 (0.67, 0.69)	<.001‡	<.001‡	0.82 (0.77, 0.86)	.45	.007
Visceral fat index + muscle area index	0.68 (0.67, 0.69)	<.001‡	<.001‡	0.82 (0.78, 0.87)	.34	.006
Visceral fat index + visceral fat density (HU)	0.68 (0.67, 0.69)	<.001‡	<.001‡	0.81 (0.77, 0.86)	.53	.04
Visceral fat index + liver PDFF	0.69 (0.68, 0.70)	<.001‡	<.001‡	0.82 (0.77, 0.86)	.40	.01
Visceral fat index + liver density (HU)	0.69 (0.68, 0.70)	<.001‡	<.001‡	0.82 (0.77, 0.86)	.43	.02
Visceral fat index + aortic calcification	0.68 (0.67, 0.69)	<.001‡	<.001‡	0.82 (0.78, 0.86)	.33	.008
Visceral fat index + muscle area index + liver PDFF	0.69 (0.68, 0.70)	<.001‡	<.001‡	0.82 (0.78, 0.87)	.33	.006
Visceral fat index + muscle area index + liver PDFF + aortic calcification	0.69 (0.68, 0.71)	<.001‡	<.001‡	0.83 (0.78, 0.87)	.22	.002

Note.—Unless otherwise indicated, data in parentheses are 95% CIs. To address the issue of multiple testing, statistical significance was assessed using the Bonferroni adjustment for the 48 separate tests conducted for each sex, with a significance threshold set at $\alpha/48$ (α divided by 48, 0.001 instead of 0.05). ADA = American Diabetes Association, AI = artificial intelligence, ASM = appendicular skeletal muscle mass, HU = Hounsfield unit, L3 = third lumbar vertebra, PDFF = proton density fat fraction, SMI = skeletal muscle index, UK = United Kingdom.

* P value compared with the reference (highest Harrell C-index among clinical prediction models).

† P value compared with the reference (highest Harrell C-index among conventional anthropometric measurements).

‡ P value remained significant after Bonferroni correction.

Discussion

The predictive ability of individual CT parameters for cardiometabolic diseases remains underexplored. In this cohort study of Korean adults, automated CT-derived body composition parameters were excellent predictors of prevalent and incident diabetes, outperforming traditional anthropometric and clinical risk models for both sexes. Visceral fat index yielded an area under the receiver operating characteristic curve of 0.70 (95% CI: 0.68, 0.71) for men and 0.82 (95% CI: 0.78, 0.85) for women for prevalent diabetes and C-index of 0.68 (95% CI: 0.67, 0.69) for men and 0.82 (95% CI: 0.77, 0.86) for women for incident diabetes. Combining CT-derived markers, including visceral fat, muscle area, liver proton density fat fraction, and aortic calcification, improved the performance of type 2 diabetes mellitus risk prediction model and accurately identified corresponding comorbidities.

In individuals with type 2 DM, initial diagnoses often coincide with comorbidities and diabetes-related complications, affecting medication choices (13). CT-derived imaging markers may facilitate a more tailored and precise diabetes treatment strategy. Building on previous research linking CT-derived body composition to diabetes diagnosis (14,25), our findings, obtained from a real-world health screening environment with regular diabetes evaluations, highlight the potential of CT imaging to enhance preventive care and risk assessment through opportunistic CT screening, as previously recommended by Pickhardt (12).

In our study, CT-derived visceral fat alone outperformed conventional type 2 DM prediction models, with its predictive performance improving when combined with other CT-derived imaging markers. This index also identified metabolic syndrome (26). Additionally, abdominal obesity, measured by the waist-to-hip ratio or waist circumference, better predicts diabetic retinopathy, diabetic kidney disease, and cardiovascular disease than does BMI (27). Given that CT is the reference standard for precisely quantifying visceral fat, an accurate assessment of visceral obesity could predict both the risk of diabetes and its complications. More than 55%–70% of individuals with type 2 DM also have metabolic dysfunction–associated steatotic liver disease (28), with a heightened risk of liver-related complications, leading to clinical guidelines advocating routine screening for this condition (13). Type 2 DM is linked to an increased risk of sarcopenia (29). In our study, patients with type 2 DM exhibited higher muscle mass but lower muscle density on CT scans, a disparity that disappeared after adjusting for BMI and age. Identification of lower muscle density, which is indicative of myosteatosis, is crucial because it is adversely associated with muscle strength and mortality (30), underscoring the need to assess both muscle mass and quality for a comprehensive muscle health evaluation.

Aortic calcification, associated with cardiovascular mortality and sharing cardiovascular disease risk factors with CAC (31), accurately identified CAC greater than 100 in our study. This potentially identifies individuals with type DM at high atherosclerotic cardiovascular disease risk (> 20 per 1000 person-years), necessitating intensive treatment initiation (21,32). Our study identified aortic calcification as a predictor of incident diabetes, suggesting that vascular calcification serves as an integrative marker of aging and overall health encompassing cardiovascular disease risk and influences diabetes onset (33,34).

In our study, CT-derived markers had a stronger association with diabetes and metabolic syndrome in women than in men, consistent with the findings of Pickhardt et al (26), who reported greater predictive accuracy in women. These sex-specific differences might be related to sexually dimorphic risk factors for cardiometabolic diseases (35). Premenopausal women, influenced by estrogen, tend to accumulate more gluteal-femoral adipose tissue, which is associated with better insulin sensitivity and a preference for lipid storage in subcutaneous rather than visceral fat (35). This estrogen-related fat distribution might explain the superior ability of CT imaging to predict diabetes risk in women (36), highlighting the importance of considering sex-specific factors and accurate fat distribution data from CT scans in risk assessment.

Our study had limitations. First, type 2 DM was diagnosed with a single measurement of fasting glucose and hemoglobin A_{1c}, diverging from typical clinical practices requiring repeat testing. However, hemoglobin A_{1c} shows reliable preanalytical stability and resistance to immediate influences such as stress and exercise, aiding in accurately determining type 2 DM incidence (37). The minimum age of individuals with prevalent and incident diabetes was 37 and 39 years, respectively, making type 1 diabetes unlikely in this age group. Second, we could not analyze pancreatic fat, a predictor of diabetes (14), suggesting a direction for future research. Third, the generalizability of our findings might be limited, as we focused on young and middle-aged Koreans, potentially not representing the broader population. Additionally, while liver US was almost universally conducted within the cohort, other tests, such as bone mineral density and CAC scans, were performed often based on participant preference, possibly introducing selection bias. However, its impact on the diagnostic utility of CT-derived image markers for diagnosing CAC or osteoporosis is expected to be minimal, given that there is no anticipated association between these conditions and CT-derived markers when participants were selecting examinations. Future research should include an unselected population to validate our findings across different demographics.

In conclusion, CT-derived parameters, particularly the visceral fat area index, outperformed traditional methods for predicting type 2 diabetes mellitus (DM) in both sexes. Combining CT-derived markers, including visceral and subcutaneous fat areas, muscle area, liver fat fraction, and aortic calcification, improved type 2 DM risk prediction performance and facilitated screening for multiple diabetes-associated comorbidities, offering tailored risk stratification. Achieving more efficient and safer approaches through reduced radiation exposure and targeted multiorgan assessments remains a necessity, and caution is warranted when considering the clinical applicability of these findings for practice.

Deputy Editor: Vicky Goh
Scientific Editor: Shannyn Wolfe (AJE)

Acknowledgment. The authors thank the other investigators, the staff, and the participants of the Kangbuk Samsung Health Study for their valuable contributions.

Author contributions: Guarantors of integrity of entire study, **Y.C., R.K., J.M.K., J.H.C., H.S.J., G.Y.L., S.R.**; study concepts/study design or data acquisition or data analysis/interpretation, all authors; manuscript drafting or manuscript revision for important intellectual content, all authors; approval of final version of submitted manuscript, all authors; agrees to ensure any questions related to the work are appropriately resolved, all authors; literature research, **Y.C., S.H.Y., J.K., Y.H.K., H.S.J., J.A.**; clinical studies, **Y.C., S.H.Y., R.K., Y.H.K., H.S.J., G.Y.L., S.R.**; statistical analysis, **Y.C., R.K., J.H.C., H.S.J., G.Y.L., S.R.**; and manuscript editing, **S.H.Y., J.M.K., H.J.C., H.S.J., S.H.W., C.D.B., S.R.**

Disclosures of conflicts of interest: Y.C. Grants from the SKKU Excellence in Research Award Research Fund, Sungkyunkwan University (2022), and the National Research Foundation of Korea, funded by the Ministry of Science, ICT, and Future Planning (NRF2021R1A2C1012626). **S.H.Y.** Stocks and stock options of MEDICAL IP. **R.K.** No relevant relationships. **J.K.** No relevant relationships. **Y.H.K.** No relevant relationships. **J.M.K.** Employee of MEDICAL IP. **H.J.C.** Employee of MEDICAL IP. **J.H.C.** No relevant relationships. **H.S.J.** No relevant relationships. **G.Y.L.** No relevant relationships. **J.A.** No relevant relationships. **S.H.W.** No relevant relationships. **C.D.B.** Research grant paid to institution from Echosens. **S.R.** No relevant relationships.

References

1. Pickhardt PJ, Summers RM, Garrett JW, et al. Opportunistic Screening: *Radiology* Scientific Expert Panel. Radiology 2023;307(5):e222044.
2. Bajwa J, Munir U, Nori A, Williams B. Artificial intelligence in healthcare: transforming the practice of medicine. Future Healthc J 2021;8(2):e188–e194.
3. Lee SJ, Graffy PM, Zea RD, Ziemlewicz TJ, Pickhardt PJ. Future Osteoporotic Fracture Risk Related to Lumbar Vertebral Trabecular Attenuation Measured at Routine Body CT. J Bone Miner Res 2018;33(5):860–867.
4. Pickhardt PJ, Pooler BD, Lauder T, del Rio AM, Bruce RJ, Binkley N. Opportunistic screening for osteoporosis using abdominal computed tomography scans obtained for other indications. Ann Intern Med 2013;158(8):588–595.
5. Graffy PM, Liu J, Pickhardt PJ, Burns JE, Yao J, Summers RM. Deep learning-based muscle segmentation and quantification at abdominal CT: application to a longitudinal adult screening cohort for sarcopenia assessment. Br J Radiol 2019;92(1100):20190327.
6. Graffy PM, Liu J, O'Connor S, Summers RM, Pickhardt PJ. Automated segmentation and quantification of aortic calcification at abdominal CT: application of a deep learning-based algorithm to a longitudinal screening cohort. Abdom Radiol (NY) 2019;44(8):2921–2928.
7. Pickhardt PJ, Graffy PM, Reeder SB, Hernando D, Li K. Quantification of Liver Fat Content With Unenhanced MDCT: Phantom and Clinical Correlation With MRI Proton Density Fat Fraction. AJR Am J Roentgenol 2018;211(3):W151–W157.
8. Lee YS, Hong N, Witanto JN, et al. Deep neural network for automatic volumetric segmentation of whole-body CT images for body composition assessment. Clin Nutr 2021;40(8):5038–5046.
9. Lee JH, Choi SH, Jung KJ, Goo JM, Yoon SH. High visceral fat attenuation and long-term mortality in a health check-up population. J Cachexia Sarcopenia Muscle 2023;14(3):1495–1507.
10. Pickhardt PJ, Graffy PM, Zea R, et al. Automated CT biomarkers for opportunistic prediction of future cardiovascular events and mortality in an asymptomatic screening population: a retrospective cohort study. Lancet Digit Health 2020;2(4):e192–e200.
11. Lee MH, Zea R, Garrett JW, Graffy PM, Summers RM, Pickhardt PJ. Abdominal CT Body Composition Thresholds Using Automated AI Tools for Predicting 10-year Adverse Outcomes. Radiology 2023;306(2):e220574.
12. Pickhardt PJ. Value-added Opportunistic CT Screening: State of the Art. Radiology 2022;303(2):241–254.
13. ElSayed NA, Aleppo G, Aroda VR, et al; on behalf of the American Diabetes Association. 4. Comprehensive Medical Evaluation and Assessment of Comorbidities: Standards of Care in Diabetes-2023. Diabetes Care 2023;46(Suppl 1):S49–S67. [Published correction appears in Diabetes Care 2023;46(9):1718-1720, 1722.]
14. Tallam H, Elton DC, Lee S, Wakim P, Pickhardt PJ, Summers RM. Fully Automated Abdominal CT Biomarkers for Type 2 Diabetes Using Deep Learning. Radiology 2022;304(1):85–95.
15. Al-Sofiani ME, Ganji SS, Kalyani RR. Body composition changes in diabetes and aging. J Diabetes Complications 2019;33(6):451–459.
16. Liu D, Garrett JW, Lee MH, Zea R, Summers RM, Pickhardt PJ. Fully automated CT-based adiposity assessment: comparison of the L1 and L3 vertebral levels for opportunistic prediction. Abdom Radiol (NY) 2023;48(2):787–795.
17. Chang Y, Kim BK, Yun KE, et al. Metabolically-healthy obesity and coronary artery calcification. J Am Coll Cardiol 2014;63(24):2679–2686.
18. Alberti KG, Eckel RH, Grundy SM, et al; International Association for the Study of Obesity. Harmonizing the metabolic syndrome: a joint interim statement of the International Diabetes Federation Task Force on Epidemiology and Prevention; National Heart, Lung, and Blood Institute; American Heart Association; World Heart Federation; International Atherosclerosis Society; and International Association for the Study of Obesity. Circulation 2009;120(16):1640–1645.
19. Janssen I, Heymsfield SB, Ross R. Low relative skeletal muscle mass (sarcopenia) in older persons is associated with functional impairment and physical disability. J Am Geriatr Soc 2002;50(5):889–896.
20. Grundy SM, Stone NJ, Bailey AL, et al. 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA Guideline on the Management of Blood Cholesterol: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. Circulation 2019;139(25):e1082–e1143.
21. Hecht HS, Blaha MJ, Kazerooni EA, et al. CAC-DRS: Coronary Artery Calcium Data and Reporting System. An expert consensus document of the Society of Cardiovascular Computed Tomography (SCCT). J Cardiovasc Comput Tomogr 2018;12(3):185–191.
22. Zou G. A modified poisson regression approach to prospective studies with binary data. Am J Epidemiol 2004;159(7):702–706.
23. Barros AJ, Hirakata VN. Alternatives for logistic regression in cross-sectional studies: an empirical comparison of models that directly estimate the prevalence ratio. BMC Med Res Methodol 2003;3(1):21.
24. Uno H, Cai T, Pencina MJ, D'Agostino RB, Wei LJ. On the C-statistics for evaluating overall adequacy of risk prediction procedures with censored survival data. Stat Med 2011;30(10):1105–1117.
25. Zou X, Zhou X, Li Y, et al. Gender-specific data-driven adiposity subtypes using deep-learning-based abdominal CT segmentation. Obesity (Silver Spring) 2023;31(6):1600–1609.
26. Pickhardt PJ, Graffy PM, Zea R, et al. Utilizing Fully Automated Abdominal CT-Based Biomarkers for Opportunistic Screening for Metabolic Syndrome in Adults Without Symptoms. AJR Am J Roentgenol 2021;216(1):85–92.
27. Wan H, Wang Y, Xiang Q, et al. Associations between abdominal obesity indices and diabetic complications: Chinese visceral adiposity index and neck circumference. Cardiovasc Diabetol 2020;19(1):118.
28. Lomonaco R, Godinez Leiva E, Bril F, et al. Advanced Liver Fibrosis Is Common in Patients With Type 2 Diabetes Followed in the Outpatient Setting: The Need for Systematic Screening. Diabetes Care 2021;44(2):399–406.
29. Liccini A, Malmstrom TK. Frailty and Sarcopenia as Predictors of Adverse Health Outcomes in Persons With Diabetes Mellitus. J Am Med Dir Assoc 2016;17(9):846–851.
30. Nachit M, Horsmans Y, Summers RM, Leclercq IA, Pickhardt PJ. AI-based CT Body Composition Identifies Myosteatosis as Key Mortality Predictor in Asymptomatic Adults. Radiology 2023;307(5):e222008.
31. Wilson PW, Kauppila LI, O'Donnell CJ, et al. Abdominal aortic calcific deposits are an important predictor of vascular morbidity and mortality. Circulation 2001;103(11):1529–1534.
32. Grundy SM, Stone NJ, Bailey AL, et al. 2018 AHA/ACC/AACVPR/AAPA/ABC/ACPM/ADA/AGS/APhA/ASPC/NLA/PCNA Guideline on the Management of Blood Cholesterol: A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. J Am Coll Cardiol 2019;73(24):e285–e350. [Published correction appears in J Am Coll Cardiol 2019;73(24):3237-3241.]
33. Shaw LJ, Raggi P, Berman DS, Callister TQ. Coronary artery calcium as a measure of biologic age. Atherosclerosis 2006;188(1):112–119.
34. Handy CE, Desai CS, Dardari ZA, et al. The Association of Coronary Artery Calcium With Noncardiovascular Disease: The Multi-Ethnic Study of Atherosclerosis. JACC Cardiovasc Imaging 2016;9(5):568–576.
35. Mauvais-Jarvis F, Bairey Merz N, Barnes PJ, et al. Sex and gender: modifiers of health, disease, and medicine. Lancet 2020;396(10250):565–582.
36. Goossens GH, Jocken JWE, Blaak EE. Sexual dimorphism in cardiometabolic health: the role of adipose tissue, muscle and liver. Nat Rev Endocrinol 2021;17(1):47–66.
37. Bonora E, Tuomilehto J. The pros and cons of diagnosing diabetes with A1C. Diabetes Care 2011;34(Suppl 2):S184–S190.